Course Code: 23MAT204

Mathematics For Intelligent Systems III

Part- 9

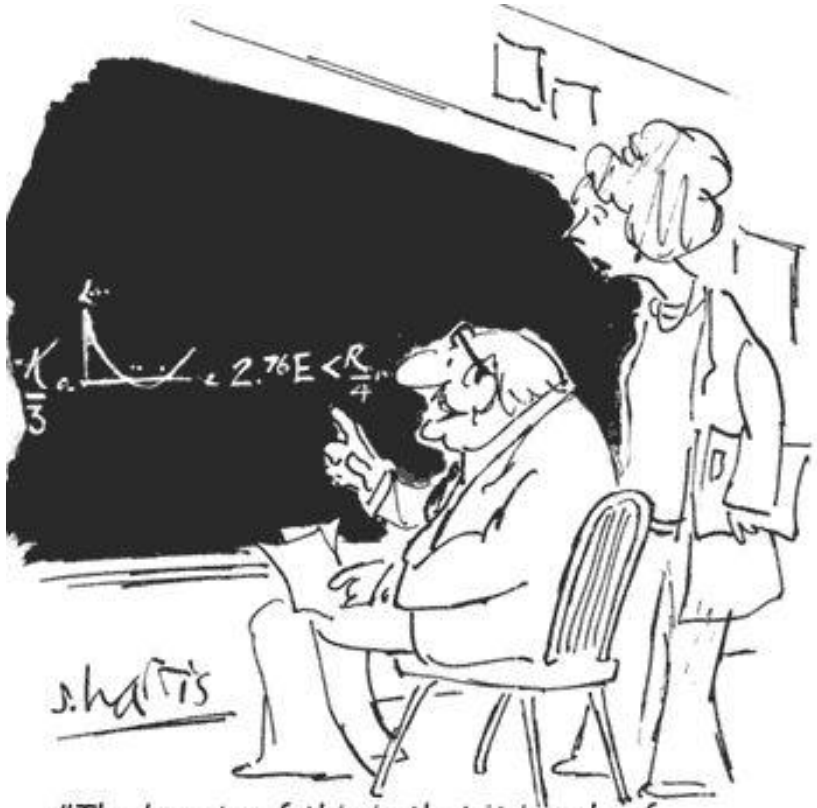
Singular Value Decomposition

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What is the use of SVD?

- SVD can be used to compute optimal low-rank approximations of arbitrary matrices.
- Face recognition
  - Represent the face images as eigenfaces and compute distance between the query face image in the principal component space
- Data mining
  - Latent Semantic Indexing for document extraction
- Image compression
  - Karhunen Loeve (KL) transform performs the best image compression
    - In MPEG, Discrete Cosine Transform (DCT) has the closest approximation to the KL transform in PSNR



"The beauty of this is that it is only of theoretical importance, and there is no way it can be of any practical use whatsoever."

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Dimensionality Reduction

- A broad class of techniques
- Goal is to compress the original representation of the input
- Example: Approximate each input $\boldsymbol{x}_n \in \mathbb{R}^D, n = 1,2, \dots, N$ as a linear combination of $K < \min\{D, N\}$ "basis" vectors $\boldsymbol{w}_1, \boldsymbol{w}_2, \dots, \boldsymbol{w}_K$, each also $\in \mathbb{R}^D$

$$\boldsymbol{x}_n \approx \sum_{k=1}^{K} z_{nk} \boldsymbol{w}_k = \mathbf{W} \boldsymbol{z}_n$$

- We have represented each $\boldsymbol{x}_n \in \mathbb{R}^D$ by a K-dim vector $\boldsymbol{z}_n$ (a new feat. rep)
- To store N such inputs $\{\boldsymbol{x}_n\}_{n=1}^N$, we need to keep $\mathbf{W}$ and $\{\boldsymbol{z}_n\}_{n=1}^N$
  - Originally we required $N \times D$ storage, now $N \times K + D \times K = (N + D) \times K$ storage
  - If $K \ll \min\{D, N\}$, this yields substantial storage saving, hence good compression

Can think of **W** as a linear mapping that transforms low-dim $\boldsymbol{z}_n$ to high-dim $\boldsymbol{x}_n$

Some dim-red techniques assume a nonlinear mapping function f such that $\boldsymbol{x}_n = f(\boldsymbol{z}_n)$

For example, f can be modeled by a kernel or a deep neural net



Note: These "basis" vectors need not necessarily be linearly independent. But for some dim. red. techniques, e.g., classic principal component analysis (PCA), they are

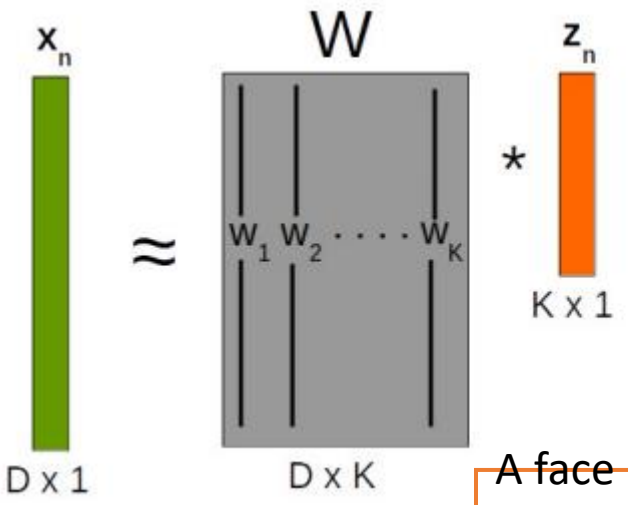
$\mathbf{W} = [\boldsymbol{w}_1, \boldsymbol{w}_2, \dots, \boldsymbol{w}_K]$ is $D \times K$

$\boldsymbol{z}_n = [z_{n1}, z_{n2}, \dots, z_{nK}]$ is $K \times 1$

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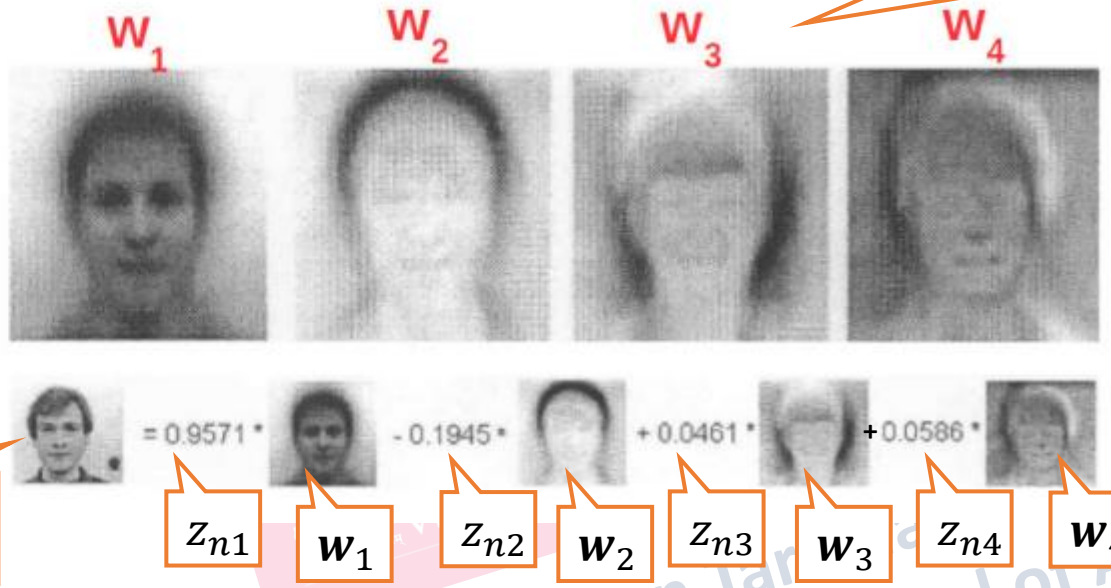
Dimensionality Reduction

- Dim-red for face images



Each "basis" image is like a "template" that captures the common properties of face images in the dataset

K=4 "basis" face images



A face image $\boldsymbol{x}_n \in \mathbb{R}^D$

z_{n1} $\boldsymbol{w}_1$ z_{n2} $\boldsymbol{w}_2$ z_{n3} $\boldsymbol{w}_3$ z_{n4} $\boldsymbol{w}_4$

- In this example, $\boldsymbol{z}_n \in \mathbb{R}^K$ $(K = 4)$ is a low-dim feature rep. for $\boldsymbol{x}_n \in \mathbb{R}^D$

Like 4 new features

- Essentially, each face image in the dataset now represented by just 4 real numbers ☺
- Different dim-red algos differ in terms of how the basis vectors are defined/learned
  - .. And in general, how the function f in the mapping $\boldsymbol{x}_n = f(\boldsymbol{z}_n)$ is defined

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Eigenvalues & Eigenvectors

- **Eigenvectors** (for a square $m \cdot m$ matrix $\mathbf{S}$)

$$\mathbf{S}\mathbf{v} = \lambda\mathbf{v}$$

(right) eigenvector $\mathbf{v} \in \mathbb{R}^m \neq \mathbf{0}$

eigenvalue $\lambda \in \mathbb{R}$

Example

$$\begin{pmatrix} 6 & -2 \\ 4 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

- How many eigenvalues are there at most?

$$\mathbf{S}\mathbf{v} = \lambda\mathbf{v} \iff (\mathbf{S} - \lambda\mathbf{I})\,\mathbf{v} = \mathbf{0}$$

only has a non-zero solution if $|\mathbf{S} - \lambda\mathbf{I}| = 0$

This is a mth order equation in λ which can have **at most m distinct solutions** (roots of the characteristic polynomial) – can be complex even though **S** is real.

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Eigen/diagonal Decomposition

- Let $\mathbf{S} \in \mathbb{R}^{m \times m}$ be a **square** matrix with ***m*** **linearly independent eigenvectors** (a "non-defective" matrix)
- **Theorem**: Exists an **eigen decomposition**

diagonal

$$\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{-1}$$

  - (cf. matrix diagonalization theorem)

Unique for distinct eigen-values

- Columns of **U** are the **eigenvectors** of **S**
- Diagonal elements of $\mathbf{\Lambda}$ are **eigenvalues** of $\mathbf{S}$

$$\mathbf{\Lambda} = \text{diag}(\lambda_1, \ldots, \lambda_m), \quad \lambda_i \geq \lambda_{i+1}$$

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Diagonal decomposition: why/how

Let **U** have the eigenvectors as columns:

$$U = \begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix}$$

Then, **SU** can be written

$$SU = S\begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix} = \begin{bmatrix} \lambda_1 v_1 & \cdots & \lambda_n v_n \end{bmatrix} = \begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix}\begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$$

Thus $\mathbf{SU} = \mathbf{U}\Lambda$, or $\mathbf{U}^{-1}\mathbf{SU} = \Lambda$

And $\mathbf{S} = \mathbf{U}\Lambda\mathbf{U}^{-1}$.

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Diagonal decomposition - example

Recall $S = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}; \lambda_1 = 1, \lambda_2 = 3.$

The eigenvectors $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ form $U = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$

Inverting, we have $U^{-1} = \begin{bmatrix} 1/2 & -1/2 \\ 1/2 & 1/2 \end{bmatrix}$

Recall $UU^{-1} = 1$.

Then, $\mathbf{S = U \Lambda U^{-1}} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1/2 & -1/2 \\ 1/2 & 1/2 \end{bmatrix}$

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Example continued

Let's divide **U** (and multiply $\mathbf{U^{-1}}$) by $\sqrt{2}$

Then, $\mathbf{S} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$

$\mathbf{Q} \not\subset (\mathbf{Q^{-1}} = \mathbf{Q^T})$

Why? Stay tuned …

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Symmetric Eigen Decomposition

- If $\mathbf{S} \in \mathbb{R}^{m \times m}$ is a **symmetric** matrix:
- **Theorem**: There exists a (unique) **eigen decomposition**
- where **Q** is **orthogonal:** $S = Q \Lambda Q^T$
  - $\mathbf{Q^{-1} = Q^T}$
  - Columns of **Q** are normalized eigenvectors
  - Columns are orthogonal.
  - (everything is real)

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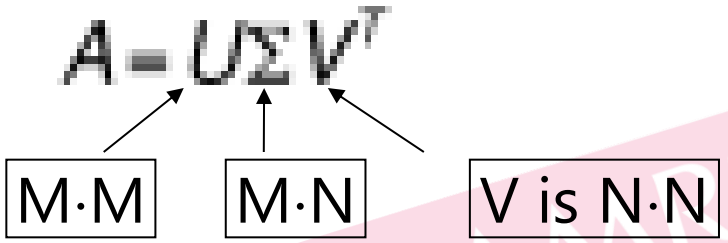
Similarity → Clustering

- We can compute the similarity between two document vector representations x_i and x_j by $x_i x_j^T$
- Let $X = [x_1 \ldots x_N]$
- Then XX^T is a matrix of similarities
- X_{ij} is symmetric
- So $XX^T = Q\Lambda Q^T$
- So we can decompose this similarity space into a set of orthonormal basis vectors (given in Q) scaled by the eigenvalues in Λ
  - This leads to PCA (Principal Components Analysis)

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Singular Value Decomposition

For an M · N matrix $\mathbf{A}$ of rank r there exists a factorization (Singular Value Decomposition = **SVD**) as follows:



(Not proven here.)

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Singular Value Decomposition

$$A = U \Sigma V^T$$

M·M M·N V is N·N

- $AA^T = Q \Lambda Q^T$
- $AA^T = (U \Sigma V^T)(U \Sigma V^T)^T = (U \Sigma V^T)(V \Sigma U^T) = U \Sigma^2 U^T$

The columns of **U** are orthogonal eigenvectors of $\mathbf{AA^T}$.

The columns of **V** are orthogonal eigenvectors of $\mathbf{A^TA}$.

Eigenvalues $\lambda_1 \ldots \lambda_r$ of $\mathbf{AA^T}$ are the eigenvalues of $\mathbf{A^TA}$.

$$\sigma_i = \sqrt{\lambda_i}$$

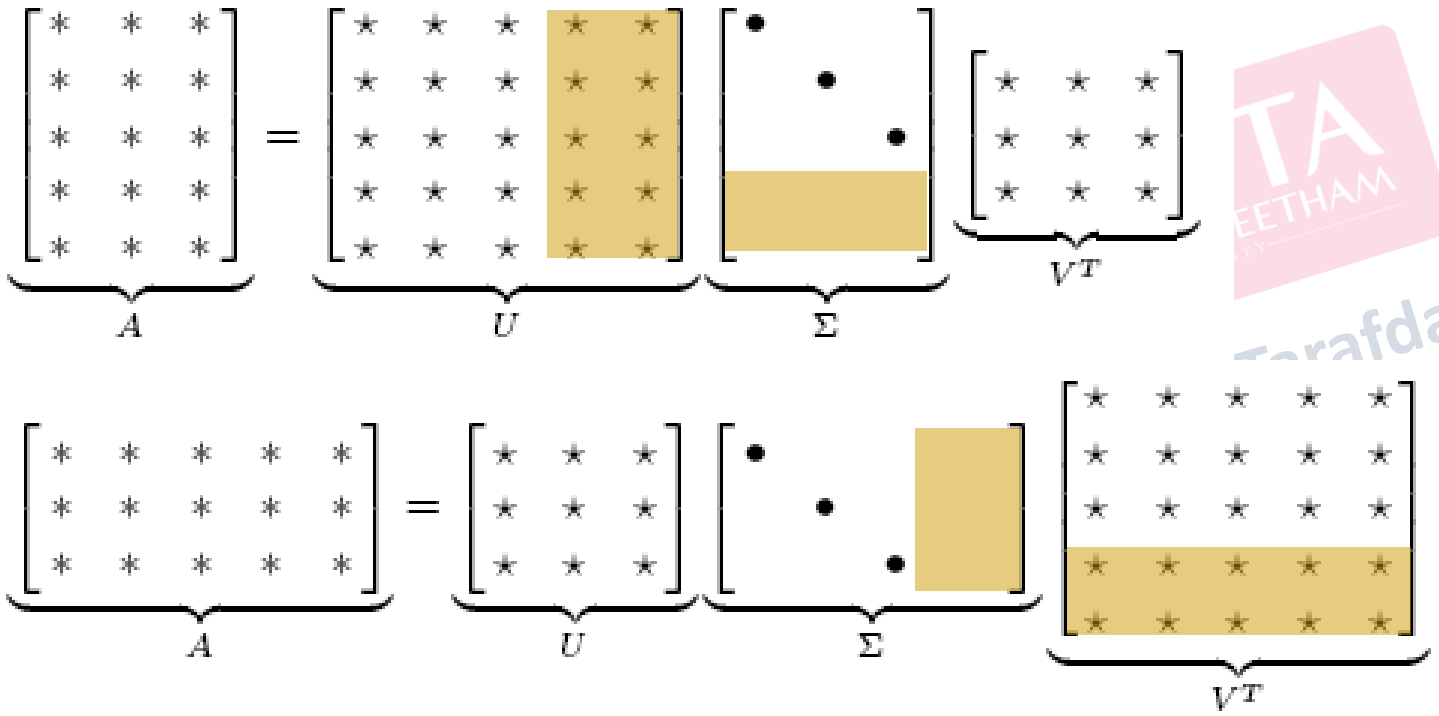
$$\Sigma = diag(\sigma_1 \ldots \sigma_r)$$

Singular values

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Singular Value Decomposition

- Illustration of SVD dimensions and sparseness



SVD example

Let $A = \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$

Thus M=3, N=2. Its SVD is

$$\begin{bmatrix} 0 & 2/\sqrt{6} & 1/\sqrt{3} \\ 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 1/\sqrt{2} & 1/\sqrt{6} & -1/\sqrt{3} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & \sqrt{3} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

Typically, the singular values arranged in decreasing order.

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Example for SVD

- Taking $\lambda^2_1=12$ and $\lambda^2_2=10$, the singular value decomposition of $\mathbf{A}$ is

$$\mathbf{A}=\begin{bmatrix}3 & 1 & 1\\ -1 & 3 & 1\end{bmatrix}$$

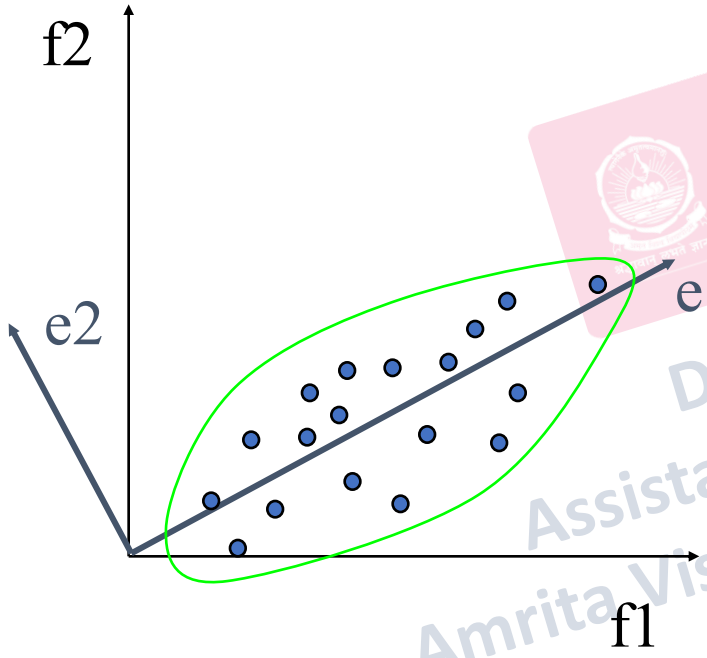
$$=\sqrt{12}\begin{bmatrix}1/\sqrt{2}\\ 1/\sqrt{2}\end{bmatrix}\begin{bmatrix}1/\sqrt{6}, 2/\sqrt{6}, 1/\sqrt{6}\end{bmatrix}+\sqrt{10}\begin{bmatrix}1/\sqrt{2}\\ -1/\sqrt{2}\end{bmatrix}\begin{bmatrix}2/\sqrt{5}, -1/\sqrt{5}, 0\end{bmatrix}$$

- Thus the U, V and Λ are computed by performing eigen decomposition of $\mathbf{AA}^T$ and $\mathbf{A}^T\mathbf{A}$
- Any matrix has a singular value decomposition but only symmetric, positive definite matrices have an eigen decomposition

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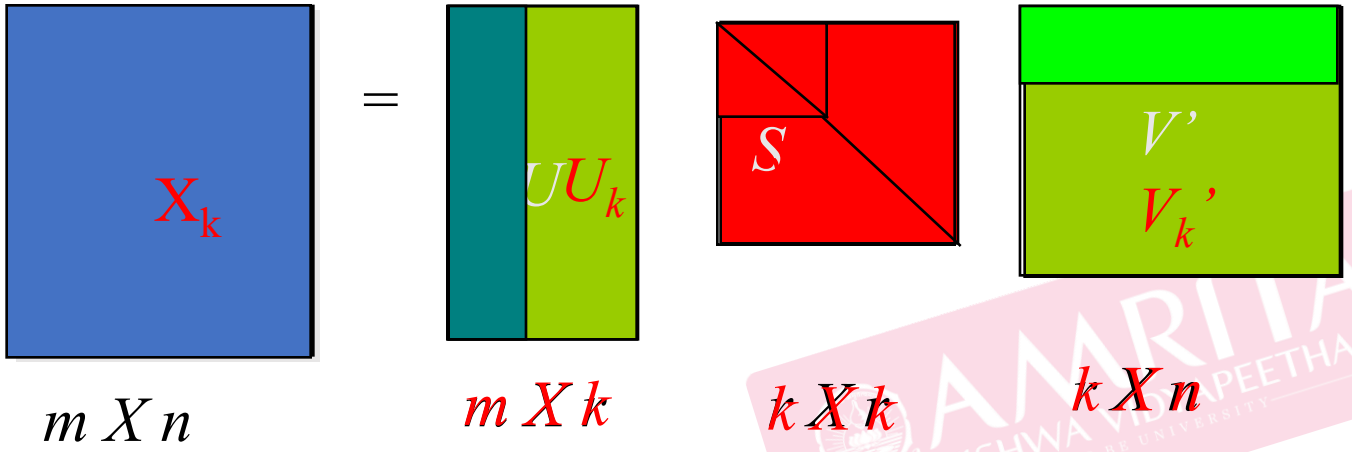
SVD

- Intuition: find the axis that shows the greatest variation, and project all points into this axis



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SVD: Mathematical Background



The reconstructed matrix $X_k = U_k.S_k.V_k'$ is the closest *rank-k* matrix to the original matrix R.

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SVD: The mathematical formulation

- Let X be the M x N matrix of M N-dimensional points
- SVD decomposition
  - $X = U \times S \times V^T$
  - U(M x M)
    - U is orthogonal: $U^TU = I$
    - columns of U are the orthogonal eigenvectors of XX^T
    - called the left singular vectors of X
  - V(N x N)
    - V is orthogonal: $V^TV = I$
    - columns of V are the orthogonal eigenvectors of X^TX
    - called the right singular vectors of X
  - *S*(M x N)
    - diagonal matrix consisting of r non-zero values in descending order
    - square root of the eigenvalues of XX^T (or X^TX)
      - r is the rank of the symmetric matrices
    - called the singular values

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SVD - Interpretation

day / customer	We 7/10/96	Th 7/11/96	Fr 7/12/96	Sa 7/13/96	Su 7/14/96
ABC Inc.	1	1	1	0	0
DEF Ltd.	2	2	2	0	0
GHI Inc.	1	1	1	0	0
KLM Co.	5	5	5	0	0
Smith	0	0	0	2	2
Johnson	0	0	0	3	3
Thompson	0	0	0	1	1

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SVD - Interpretation

- **X** = **U** **S** $\mathbf{V}^T$ - example:

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 2 & 2 & 2 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0.18 & 0 \\ 0.36 & 0 \\ 0.18 & 0 \\ 0.90 & 0 \\ 0 & 0.53 \\ 0 & 0.80 \\ 0 & 0.27 \end{bmatrix} \times \begin{bmatrix} 9.64 & 0 \\ 0 & 5.29 \end{bmatrix} \times \begin{bmatrix} 0.58 & 0.58 & 0.58 & 0 & 0 \\ 0 & 0 & 0 & 0.71 & 0.71 \end{bmatrix}$$

v1

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SVD - Interpretation

- **X** = **U** S $\mathbf{V}^T$ - example:

variance ('spread') on the v1 axis

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 2 & 2 & 2 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0.18 & 0 \\ 0.36 & 0 \\ 0.18 & 0 \\ 0.90 & 0 \\ 0 & 0.53 \\ 0 & 0.80 \\ 0 & 0.27 \end{bmatrix} \times \begin{bmatrix} 9.64 & 0 \\ 0 & 5.29 \end{bmatrix} \times \begin{bmatrix} 0.58 & 0.58 & 0.58 & 0 & 0 \\ 0 & 0 & 0 & 0.71 & 0.71 \end{bmatrix}$$

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SVD - Interpretation

- $\mathbf{X} = \mathbf{U}\,\mathbf{S}\,\mathbf{V}^T$ - example:
  - $\mathbf{U}\,\Lambda$ gives the coordinates of the points in the projection axis

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 2 & 2 & 2 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0.18 & 0 \\ 0.36 & 0 \\ 0.18 & 0 \\ 0.90 & 0 \\ 0 & 0.53 \\ 0 & 0.80 \\ 0 & 0.27 \end{bmatrix} \text{x} \begin{bmatrix} 9.64 & 0 \\ 0 & 5.29 \end{bmatrix} \text{x} \begin{bmatrix} 0.58 & 0.58 & 0.58 & 0 & 0 \\ 0 & 0 & 0 & 0.71 & 0.71 \end{bmatrix}$$

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Dimensionality reduction

- set the smallest eigenvalues to zero:

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 2 & 2 & 2 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0.18 & 0 \\ 0.36 & 0 \\ 0.18 & 0 \\ 0.90 & 0 \\ 0 & 0.53 \\ 0 & 0.80 \\ 0 & 0.27 \end{bmatrix} \times \begin{bmatrix} 9.64 & 0 \\ 0 & 5.29 \end{bmatrix} \times \begin{bmatrix} 0.58 & 0.58 & 0.58 & 0 & 0 \\ 0 & 0 & 0 & 0.71 & 0.71 \end{bmatrix}$$

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Dimensionality reduction

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 2 & 2 & 2 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 0.18 & 0 \\ 0.36 & 0 \\ 0.18 & 0 \\ 0.90 & 0 \\ 0 & 0.53 \\ 0 & 0.80 \\ 0 & 0.27 \end{bmatrix} \times \begin{bmatrix} 9.64 & 0 \\ 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0.58 & 0.58 & 0.58 & 0 & 0 \\ 0 & 0 & 0 & 0.71 & 0.71 \end{bmatrix}$$

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Dimensionality reduction

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 2 & 2 & 2 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 0.18 & 0 \\ 0.36 & 0 \\ 0.18 & 0 \\ 0.90 & 0 \\ 0 & 0.53 \\ 0 & 0.80 \\ 0 & 0.27 \end{bmatrix} \times \begin{bmatrix} 9.64 & 0 \\ 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0.58 & 0.58 & 0.58 & 0 & 0 \\ 0 & 0 & 0 & 0.71 & 0.71 \end{bmatrix}$$

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Dimensionality reduction

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Dimensionality reduction

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 2 & 2 & 2 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 2 & 2 & 2 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

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Dimensionality reduction

Equivalent:

'spectral decomposition' of the matrix:

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 2 & 2 & 2 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0.18 & 0 \\ 0.36 & 0 \\ 0.18 & 0 \\ 0.90 & 0 \\ 0 & 0.53 \\ 0 & 0.80 \\ 0 & 0.27 \end{bmatrix} \text{x} \begin{bmatrix} 9.64 & 0 \\ 0 & 5.29 \end{bmatrix} \text{x} \begin{bmatrix} 0.58 & 0.58 & 0.58 & 0 & 0 \\ 0 & 0 & 0 & 0.71 & 0.71 \end{bmatrix}$$

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Dimensionality reduction

Equivalent:

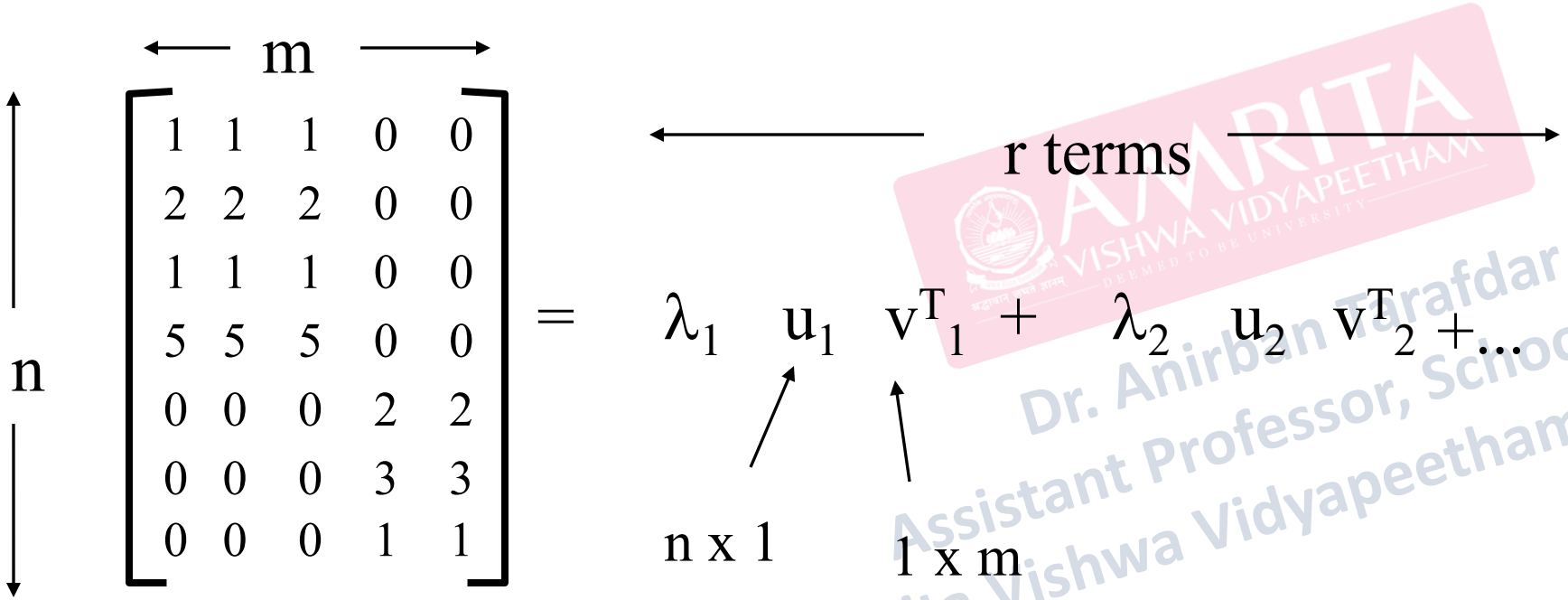
'spectral decomposition' of the matrix:

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 2 & 2 & 2 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} | & | \\ u_1 & u_2 \\ | & | \end{bmatrix} \times \begin{bmatrix} \lambda_1 & \varnothing \\ \varnothing & \lambda_2 \end{bmatrix} \times \begin{bmatrix} — & v_1 & — \\ — & v_2 & — \end{bmatrix}$$

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Dimensionality reduction

‘spectral decomposition’ of the matrix:

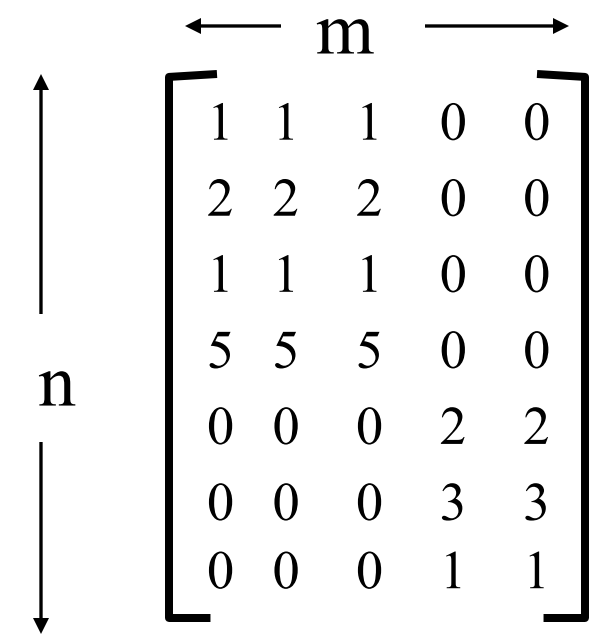


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Dimensionality reduction

approximation / dim. reduction:

by keeping the first few terms (Q: how many?)



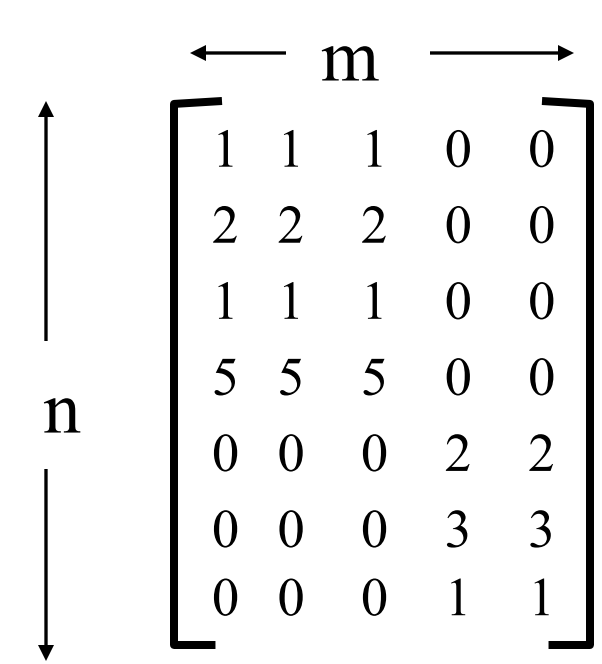
$$= \lambda_1 \; u_1 \; v^T_1 + \lambda_2 \; u_2 \; v^T_2 + \ldots$$

assume: $\lambda_1 >= \lambda_2 >= \ldots$

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Dimensionality reduction

A heuristic: keep 80-90% of 'energy' (= sum of squares of λ_i's)



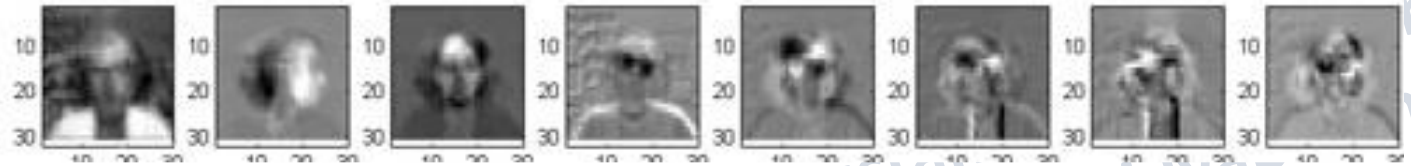
$$= \lambda_1 \; u_1 \; v^T_1 + \lambda_2 \; u_2 \; v^T_2 + \ldots$$

assume: $\lambda_1 >= \lambda_2 >= \ldots$

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Another example-Eigenface

- The PCA problem in HW5
- Face data X
- Eigenvectors associated with the first few large eigenvalues of XX^T have face-like images



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Dimensionality reduction

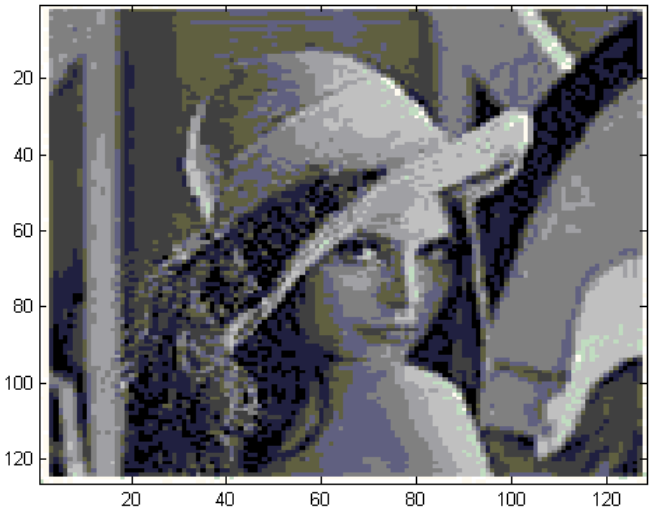
- Matrix V in the SVD decomposition ($X = USV^T$) is used to transform the data.
- XV (= US) defines the transformed dataset.
- For a new data element x, xV defines the transformed data.
- Keeping the first k ($k < n$) dimensions, amounts to keeping only the first k columns of V.

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Taking Care of Business

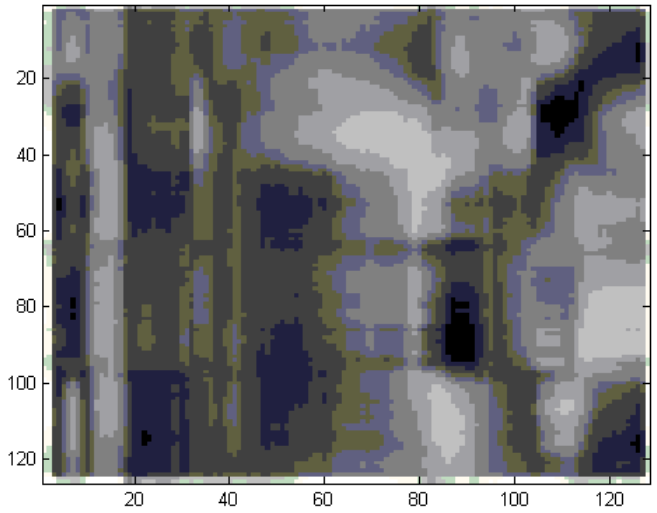
- SVD, singular values = rank(A)
- $A = \sigma_1 u_1 v^T_1 + \ldots \sigma_k u_k v^T_k + 0 * u_{k+1} v^T_{k+1}$
- Approximate A by eliminating small singular values

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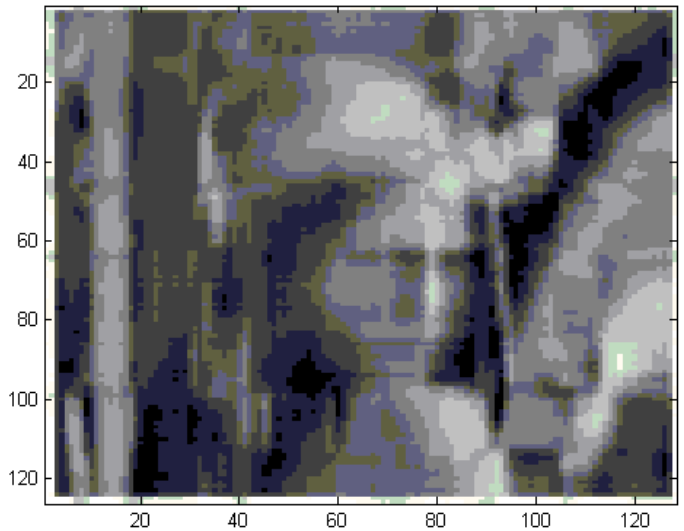


The original,
k=126

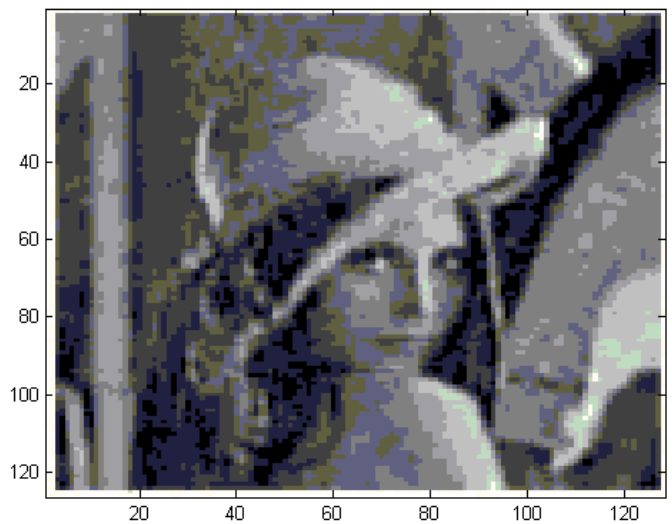
k=4



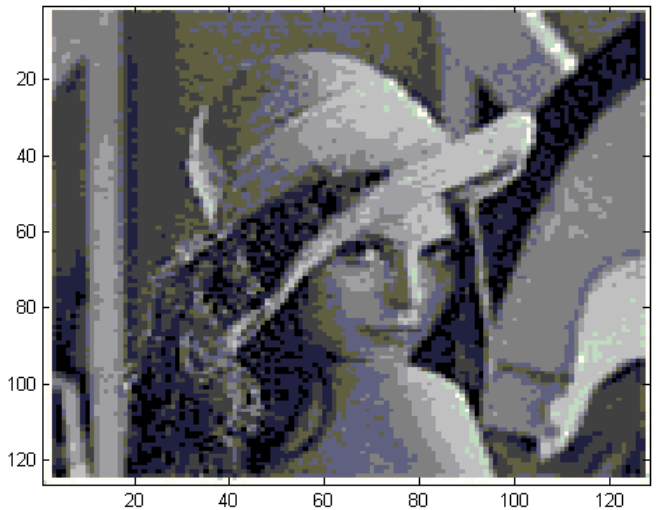
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k=20

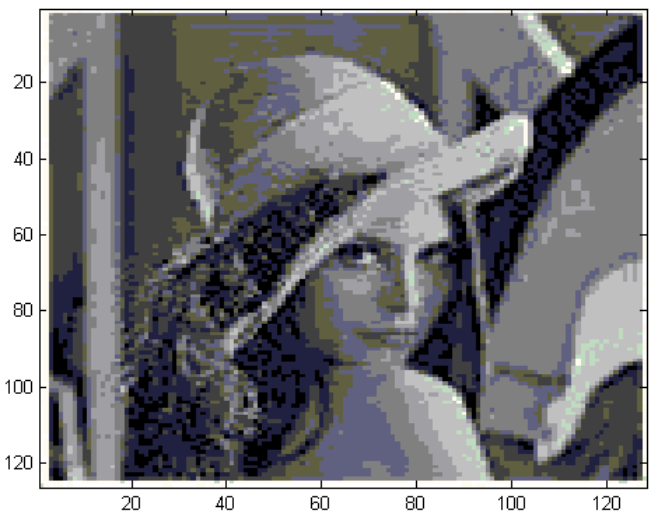


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k=50

Original,
again,
k=126



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Normal vs Noised Image

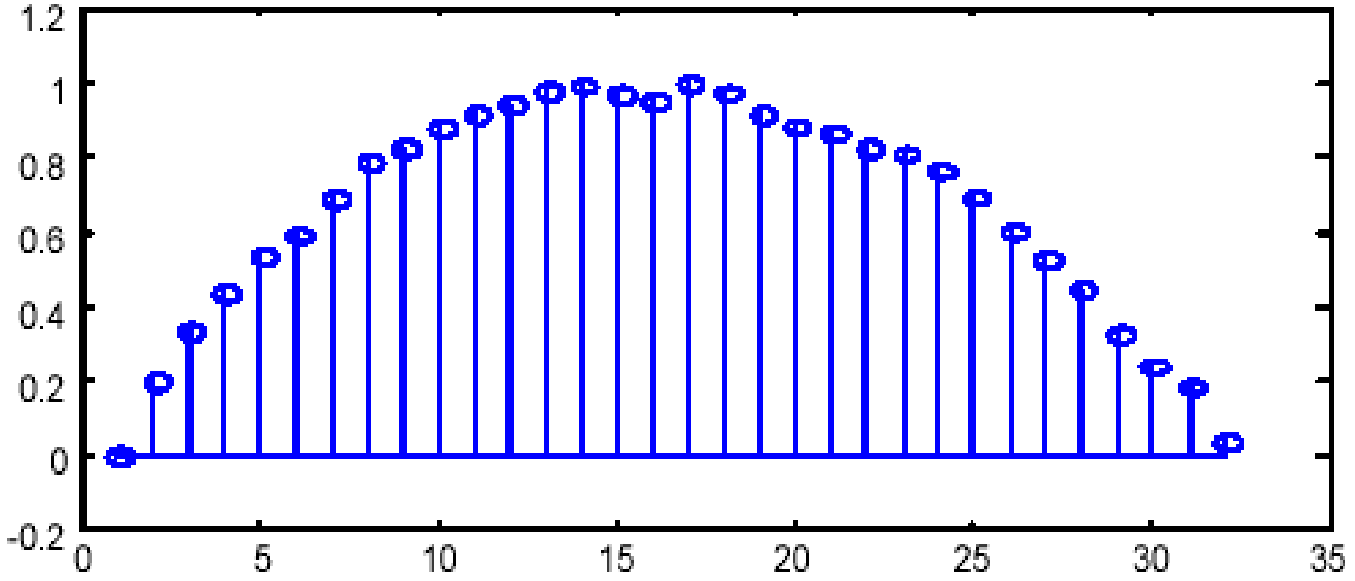


Figure 1: *The normalized difference between singular values of noised and original image, averaged for 256x256 image partitioned in blocks of size b=32.*

Block SVD filtering algorithm

- Let the original, non-corrupted image F be represented as KxL matrix. Adding the noise to the original F image will produces the noised image G of the same size
  - **G = F + N**
- Where N = random KxL noise field
- G will be divided into bxb matrix image
- $G_{ij} = U_{ij} S_{ij} V_{ij}^T$ where i = 1,2,3,4,…k; j =1,2,3,4,…,l
- Where U_{ij} is the left singular vector, S_{ij} is the diagonal, and V_{ij} is the right singular vector
- From here we can calculate the mean value of the rank (rank obtained from $G_{ij} * S_{ij}$)

$$\bar{r}_F = \frac{1}{k\,l} \sum_{i=1}^{k} \sum_{j=1}^{l} rank\,(\mathbf{F}_{i\,j})$$

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SVD's calculation

- The proposed algorithm has two basic steps: first, the noise variance (**variance** of a random variable (or somewhat more precisely, of a probability distribution) is a measure of its statistical dispersion, indicating how its possible values are spread around the expected value) is estimated,
- And then filtering is performed on singular values and vectors. Noised image is divided into square blocks of size *bxb*. For each block the singular value decomposition is performed. In the consequent step, the average sum of the last *t* singular values is calculated over all image

$$n_s = \frac{1}{k\,l} \sum_{i=1}^{k} \sum_{j=1}^{l} \sum_{r=b-t+1}^{b} S_{ijr}$$

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